Nykaa Online Marketing Campaign Analysis (July-2024 to June 2025)

Objective - To analyze impact of online campaign performed by different campaign channels on different customer segments and solving most of the business problems related to Campaign Funnel, Campaign performance, Customer Segmentation, Revenue and help business in taking data driven decisions.

About Dataset -This dataset simulates digital marketing campaign data from an e-commerce platform. Each row represents campaign performance metrics tracked over specific marketing channels.

Enable Business to:

- Analyze high-performing vs low-performing campaigns
- Calculate Revenue and Forecasting Revenue.
- Detect patterns in customer engagement Campaign funnel
- Customer segmentation against Revenue

Approach Used

1	Data wrangling
2	Data Modeling
3	Feature engineering
4	Statical Modeling
5	Interpretation
6	Visualization
7	Recommendations

Tools Used

1	MS Excel
2	Power BI

Query: 1

Conduct statical modeling on campaign and check performance of different campaign type in account of Leads. Which online campaign is better among all when it comes to generating leads?

Solution: 1

We have 5 different campaign type including – Emails, Influencer, Paid Ads, SEO and Social Media. Let's perform ANOVA – Single factor analysis (In MS Excel) to check which campaign is better among all.

Anova: Single Factor						
SUMMARY						
Groups	***Count***	***Sum***	***Average***	***Variance***		
Email	11112	20721437	1864.780	2058885.534		
Influencer	11134	20951392	1881.749	2003345.971		
Paid Ads	11116	20778836	1869.273	2046749.373		
SEO	11079	20763366	1874.119	2080301.794		
Social Media	11114	21076766	1896.416	2116264.605		
ANOVA						
Source of Variation	***SS***	***df***	***MS***	***F***	***P-value***	***F crit***
Between Groups	6851705.086	4	1712926.271	0.831083455	0.505092	2.372092
Within Groups	1.14493E+11	55550	2061076.129			
Total	1.145E+11	55554				

Interpretation:

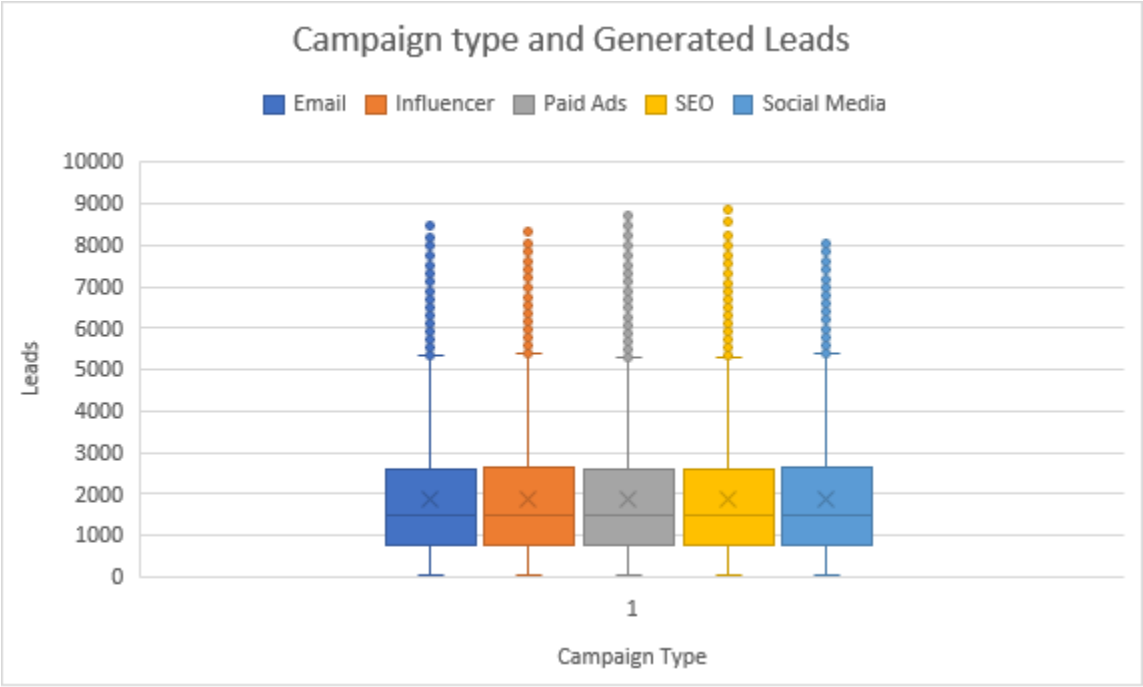
Look at the averages of each group, There is no significant difference in number of Leads among each campaign.

ANOVA Results:

Using P-value: 0.505092 > 0.05 Means Not statically significant

Using F-Test: 0.83<2.37 F crit > F Means Not Significant

Using Box plot for this Visualization



Conclusion: All channels (Email, Influencer, Paid Ads, SEO, Social-Media) perform almost equally. Channel selection does not impact this campaign significantly.

Query 2:

Does increase in Leads really leads to increase in revenue? Forecast revenue with respect to leads.

Solution: 1

Let's check if Leads really drives the revenue, I'll conduct a Linear regression to check whether Leads is a business lever in generating the revenue.

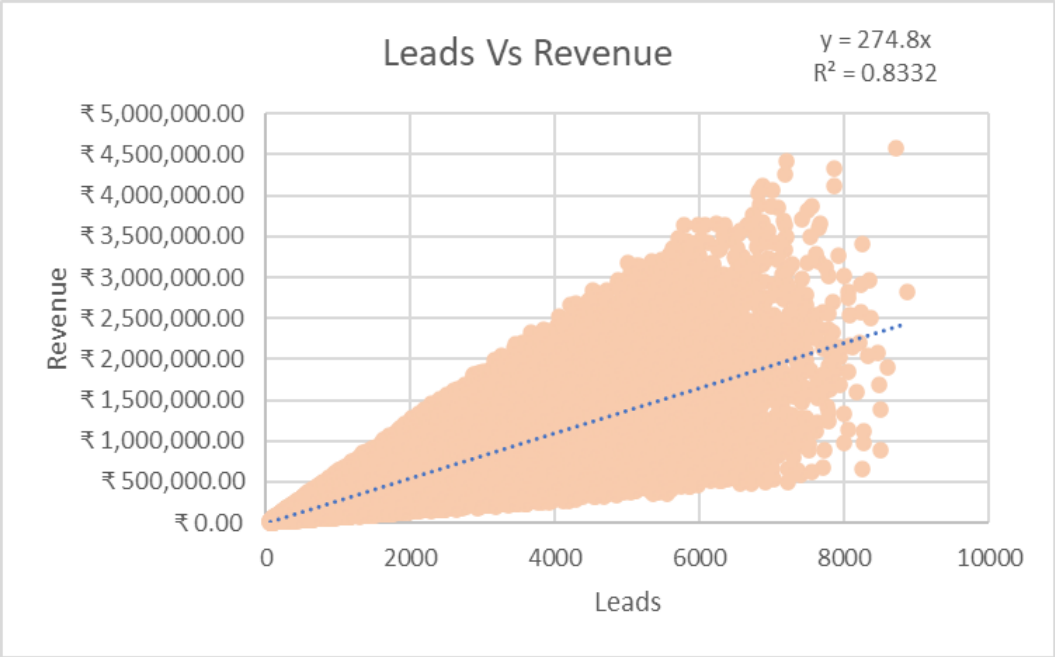
Regression Statistics						
Multiple R	0.805234646					
R Square	0.648402834					
Adjusted R Square	0.648396505					
Standard Error	290558.3807					
Observations	55555					
ANOVA						
	df	***SS***	***MS***	***F***	***Significance F***	
Regression	1	8.64916E+15	8.64916E+15	102448.843	0	
Residual	55553	4.69002E+15	84424172610			
Total	55554	1.33392E+16				
	Coefficients	***Standard Error***	***t Stat***	***P-value***	***Lower 95%***	***Upper 95%***
Intercept	-135.3618	2029.313782	-0.066703265	0.94681818	-4112.83044	3842.10673
Leads	274.8431337	0.858680032	320.0763074	0	273.1601151	276.5261523

Interpretation:

Multiple R (Coefficient) = 0.805234646 Means There is a strong positive relationship between Leads and Revenue.
R Square = 0.648402834 Means About 64.8% of the variation in Revenue is explained by Leads (Good Model Fit)
Adjusted R Square = 0.648396505, The model is stable and reliable.

The model is **highly statistically significant**. P value of Leads is 0, Means Lead is statically significant when it comes to generating revenue.
Leads is a business lever and has significant relationship with Revenue.

Using Scatter plot for this Visualization



Conclusion: Leads are a strong and significant predictor of revenue, explaining nearly 65% of its variation. Increasing lead generation will directly and substantially increase revenue.

Revenue Forecasting against the Leads

Each additional lead increases revenue by **₹274.84.** Maximum lead is 8876, Let's add more leads (Mode which is 636) to it in **Ascending Manner** to predict Revenue.

If Leads	Then Revenue
9512	₹2,614,171.26
10148	₹2,788,971.40
10784	₹2,963,771.55
11420	₹3,138,571.70
12056	₹3,313,371.85
12692	₹3,488,172.00
13328	₹3,662,972.14
13964	₹3,837,772.29

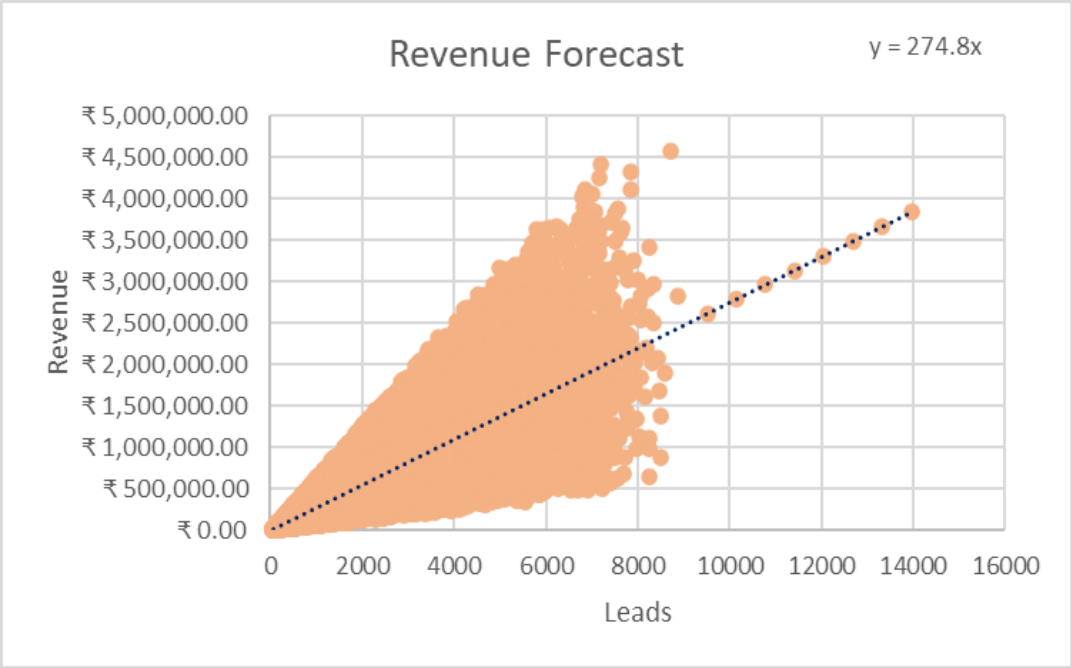
Let's say we want to generate 42 Lacs revenue, how much leads we supposed to generate?

Required Leads = Revenue Target/Revenue per lead

Required Leads = 4200000/274.84 = 15,287 leads.

We need approximately 15300 leads to generate revenue of 42 lacs.

Using Scatter plot for Revenue Forecasting Visualization



Conclusion: The forecasted revenue values are consistent with the regression model, indicating a linear relationship between leads and revenue. The more leads we generate more revenue we get. We can set a Revenue target and chase for specific number of leads to achieve that Revenue Target.

Query 3:

Company was specifically targeting 5 customer segments in this online campaign including College Students, Premium Shoppers, Tier 2 City Customers, Working Women and youth. Check which customer segment is generating more revenue and which customer segment is generating less revenue, Also advice strategy to pull up customer segment with less revenue.

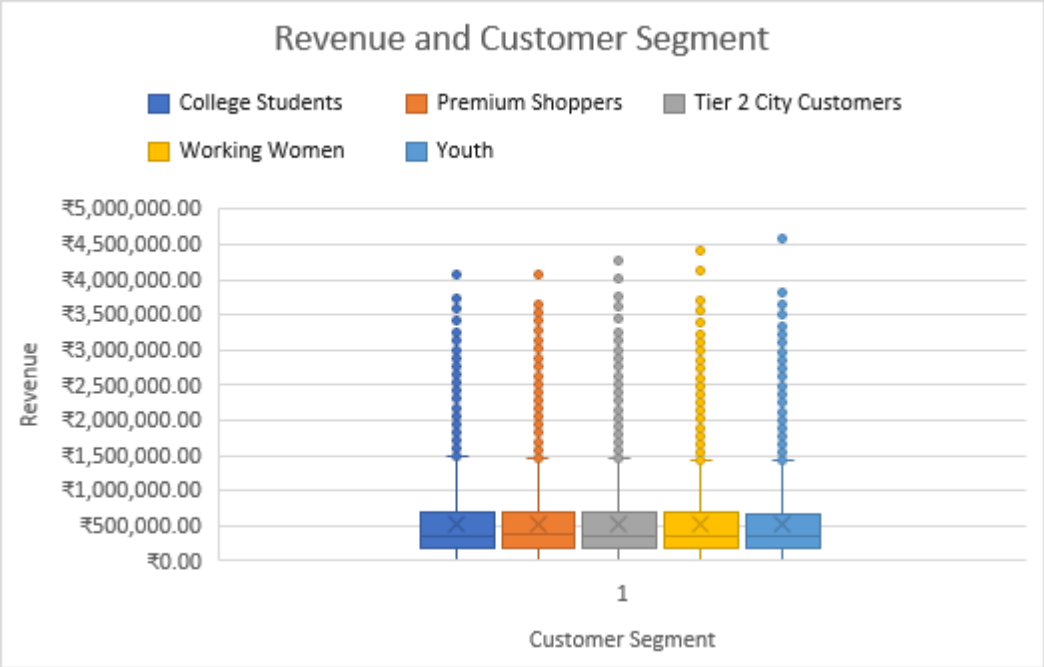
Solution 3:

Let's conduct ANOVA one Factor analysis to find out customer segment is generating more revenue and which customer segment is generating less revenue.

Anova: Single Factor						
SUMMARY						
Groups	***Count***	***Sum***	***Average***	***Variance***		
College Students	11078	5786247365	522318.7728	2.46976E+11		
Premium Shoppers	11182	5852284945	523366.5664	2.45039E+11		
Tier 2 City Customers	11100	5701063841	513609.3550	2.36474E+11		
Working Women	11224	5743709729	511734.6516	2.37983E+11		
Youth	10971	5573058402	507980.8953	2.33921E+11		
ANOVA						
Source of Variation	***SS***	***df***	***MS***	***F***	***P-value***	***F crit***
Between Groups	2.02E+12	4	5.05113E+11	2.103822795	0.077512	2.372092
Within Groups	1.33E+16	55550	2.40093E+11			
Total	1.33E+16	55554				

As per above model and after analyzing p-Value, f and f-crtical, This is very clear that every customer segment is generating almost equal revenue there is not statical differece in the revenue generated by College Students, Premium Shoppers, Tier 2 City Customers, Working Women and youth. Let's use Box plot for visualization.

Using Box plot for this Visualization



Conclusion: Box plot also support the ANOVA test that there is not significant difference in Revenue among College Students, Premium Shoppers, Tier 2 City Customers, Working Women and youth. But premium shoppers and working women is having scope of improvement, We can improve our product/Positioning for these two segments.

Since, Largely there is no segment with less Revenue, We should make strategy within the Segment including - Optimize funnels (landing pages, forms), Personalized messaging, Retargeting campaigns and Better creatives.

Query 4:

Does engagement score which is assigned as per the engagement performance has correlation with Revenue? Is high engagement score means high revenue?

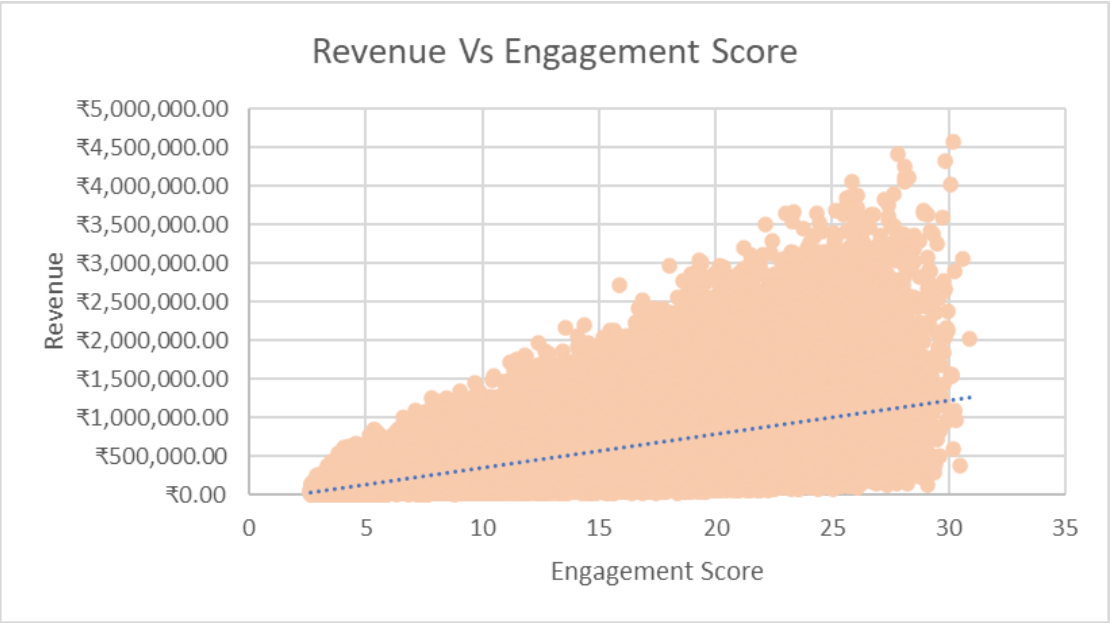
Solution 4:

Let's conduct Coefficient correlation model for this analysis.

Correlation Analysis		
	Engagement_Score	*Revenue*
Engagement_Score	1	
Revenue	**0.564719088**	1
R = 0.56471		

Since **R = 0.56471, This shows Moderate positive correlation, Increasing engagement will likely increase revenue, but the effect is not as powerful as leads.**

Using Scatter plot for this Visualization



Conclusion: This suggests that as engagement increases, revenue tends to increase, although the relationship is not very strong and other factors may also influence revenue. Rather Use engagement to drive leads and use leads to drive revenue.

Query 5:

Conduct a test and figure out Campaign funnel including CTR, Drop off % and check for leakage. Provide your recommendation.

Solution:

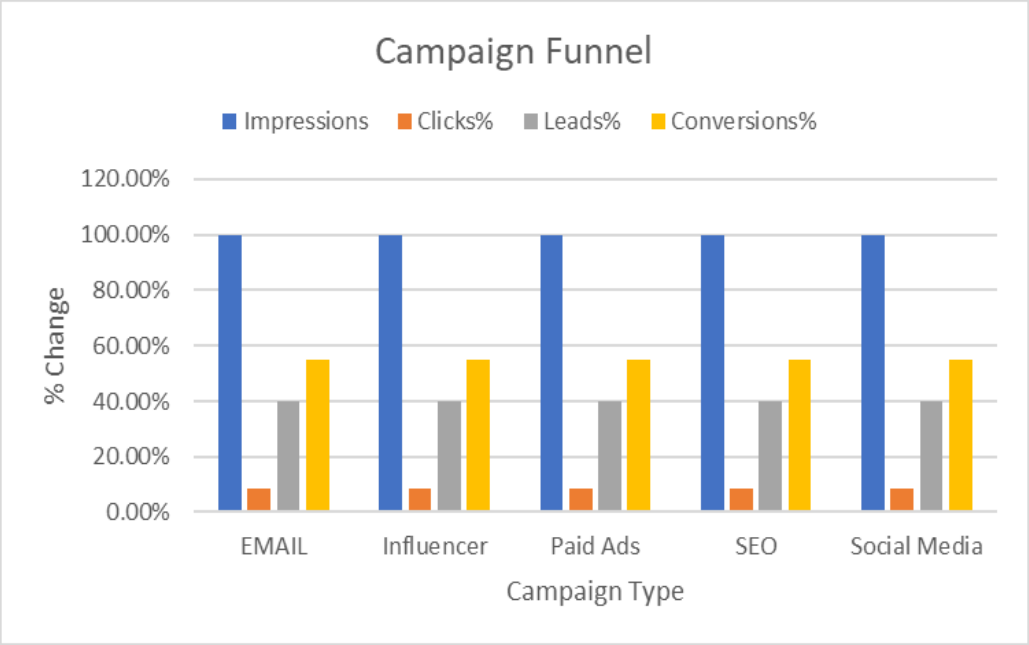
Let's arrange data and analyze through Pivot data analysis model.

Campaign_Type	Impressions	Clicks	Leads	Conversions
EMAIL	608489897	51664577	20721437	11425766
Influencer	613352995	52420253	20942669	11559308
Paid Ads	613013176	52188231	20775273	11423038
SEO	610321918	51608334	20761105	11368842
Social Media	614603708	52512670	21073499	11594210

Campaign_Type	Impressions%	Clicks%	Leads%	Conversions%
EMAIL	100.00%	8.49%	40.11%	55.14%
Influencer	100.00%	8.55%	39.95%	55.20%
Paid Ads	100.00%	8.51%	39.81%	54.98%
SEO	100.00%	8.46%	40.23%	54.76%
Social Media	100.00%	8.54%	40.13%	55.02%

- Impressions → Click, CTR is Approximately 8.50% which is decent but could go at least @ 10%
- Click → Lead Rate is Approximately 40%, Means landing page is working well and 40% is strong indication.
- Lead → Conversion Rate is Approximately 55%, Which is very good indication that Sales/Operations team is working efficiently.
- Overall Funnel Efficiency @ 1.87%

Using Clustered Column for this Visualization



Recommendations

-Our CTR is around 8.5% which is decent but It's a weak point as well (losing most users at the awareness stage)

-Our strength is conversion engine.

-We should optimize Ads: Better creatives, Better targeting, Strong CTA.

-Feed it more traffic

-Target Working